CONTENT PORTFOLIO

Adithya Vikram

AI-assisted science & maths content · Class 6–8 and Class 11–12 physics

https://drive.google.com/drive/folders/10NAg8tTV5YEnMx2jsSIvik3hg6fnfYNA

Class 6–8 samples

made for this role

VIDEO · SCIENCE · CLASS 8

Why does a sharp knife cut? Pressure = force ÷ area

An animated lesson (about 2 min): a puzzle hook, a brick drawn to scale, the rule, worked numbers, a knife and school-bag example, a pause-and-try question and a recap. Every size and number is calculated by code.

0. Class 6–8 samples (folder)

PDF · MATHS · CLASS 6–7

Fractions: notes, worksheet and answer key

Two pages of picture-based notes, a 14-question worksheet in 5 sections (from easy to word problems) and a full answer key with working. All answers are computed and double-checked by code.

0. Class 6–8 samples (folder)

Higher classes: physics

VIDEO · OVERVIEW

Start Here: how I can help (2:36)

How I combine AI and code to make checked content, with real examples. Voiced in my own (AI-cloned) voice.

1. Start Here

VIDEO · JEE

Work done on a rough dome: full lesson (10 min)

A hard JEE question, explained step by step from zero: forces, splitting the weight, the integral, a second check with energy, and the common traps. In my own (AI-cloned) voice.

3. Problem Solving

VIDEO · 9:16

Short: AI + Physics, checked by code

A 44-second motion-graphics Short for YouTube and Reels, in my own (AI-cloned) voice.

5. Short

PDF · CLASS 11 · JEE

Rough-dome problem: class notes

Four pages that match the video: diagrams drawn to scale, every step, the energy check, common traps, and 5 practice questions with answers checked by code.

4. Class Notes

AI tools I used, and how

Tool	What I use it for
Claude (Anthropic), an AI assistant	Drafting scripts, captions and explanations, and writing the animation and checking code. I decide the topic, the teaching order and the examples, and I review every script and frame.
Code that checks the maths	Every number is calculated by code, never typed in by hand or taken from AI text. Each answer is checked a second, independent way (for example a simulation, or a different unit route). Right now **297 of 297** automatic checks pass.
Remotion (video made with code)	Animations where every frame comes from the equations, so motion and sizes are physically correct. One scene can come out as a 16:9 lesson, a 9:16 Short, or in another language.
KaTeX	Clean, correct maths formulas on screen and in PDFs.
Text-to-speech	Every video is voiced by an AI copy of **my own voice**, made with Chatterbox (free, open source, runs on my laptop) from a short recording of me, then cleaned of background noise and levelled. A check re-records any line that comes out rushed.

My rule: AI for words, code for numbers

AI helps with
scripts, captions, simpler wording, Hinglish versions, hooks, new story settings for old questions

Code decides
every number, every graph, every frame of motion, and the second check of each answer

If an answer can't be checked, it is marked **"needs teacher review"** and never shown as final.

How one piece gets made

1. **Plan:** pick the topic, the student level and one clear example.
2. **Write:** draft the script with AI help, then rewrite it in plain, spoken language.
3. **Calculate:** code works out every number and checks it a second way.
4. **Animate and render:** video, Short, notes or worksheet, in English or Hinglish.
5. **Review:** I watch or read everything before it goes out.

All questions, scripts, visuals and the mascot (Kan) are original. Thank you for reviewing my work. Adithya Vikram